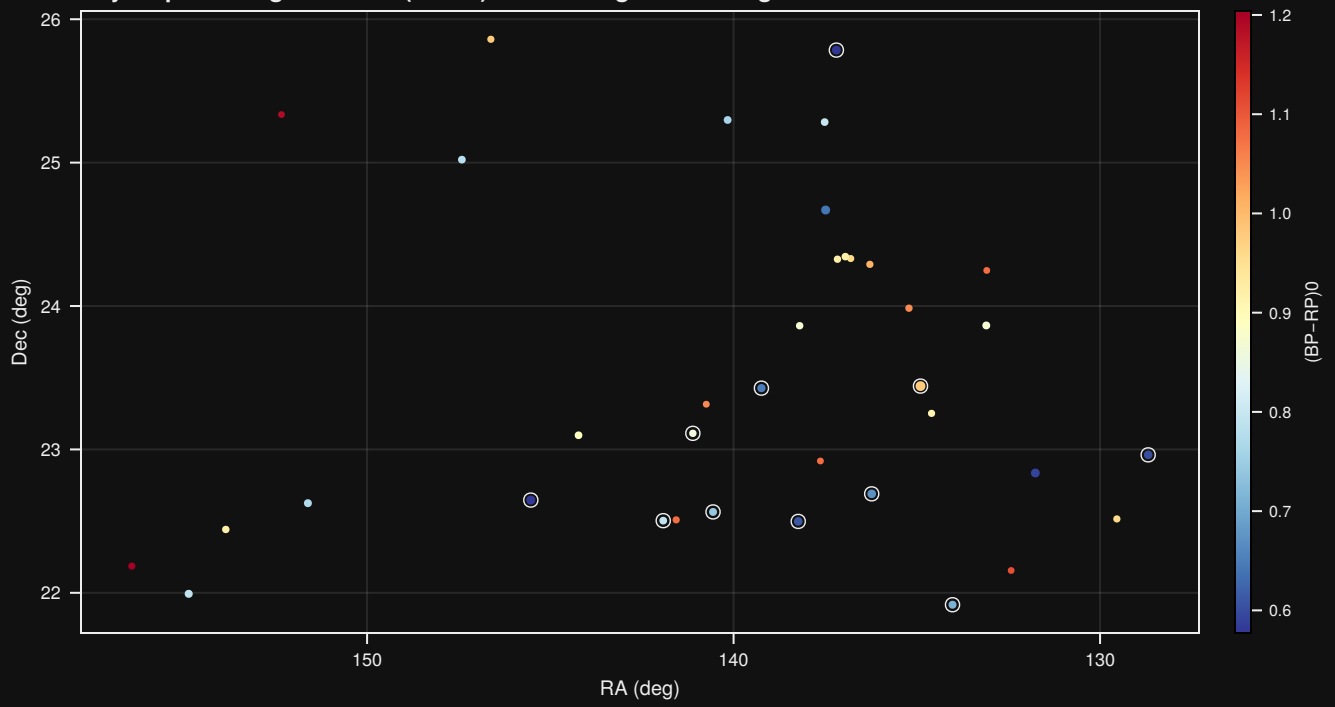
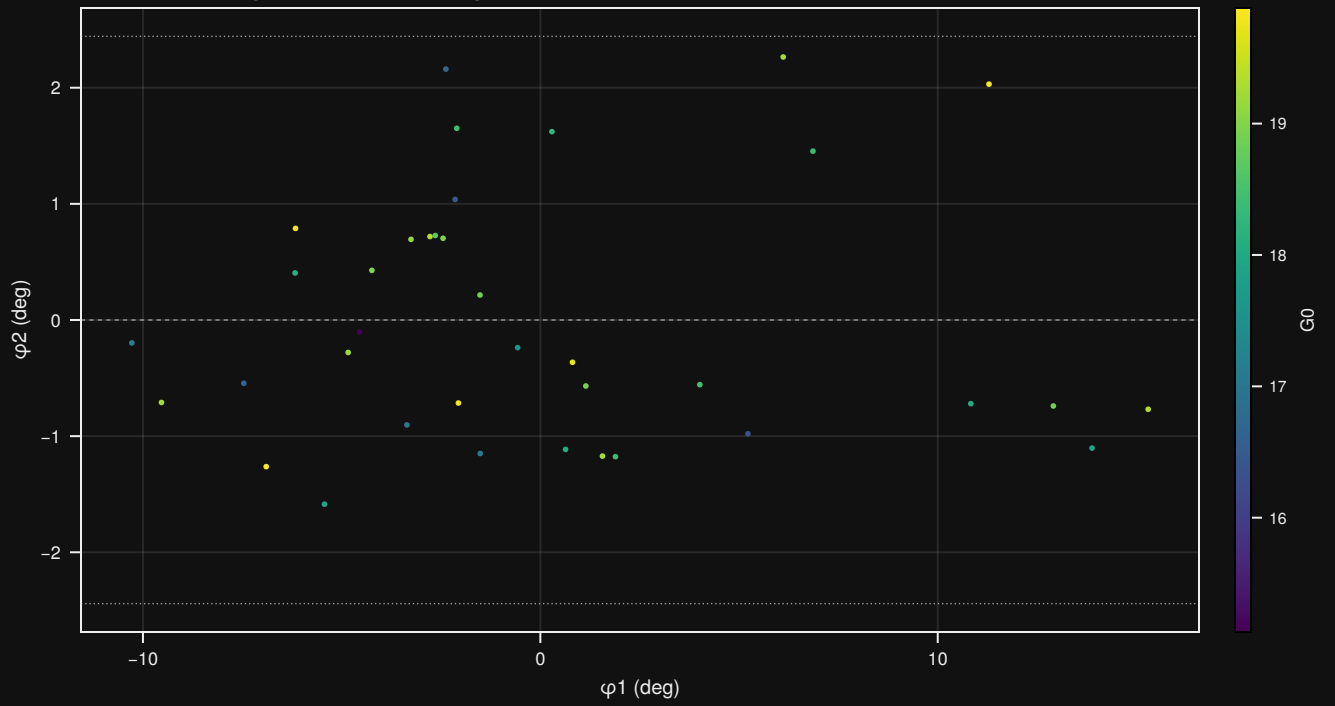
New-7

MEMBERSHIP	
Stars in catalog	37
with Gaia GSP-Phot	22 (59%)
with radial velocity	11
Stream id (sID)	27
GEOMETRY · great-circle fit	
Centroid RA, Dec	139.9, 23.3 deg
Centroid l, b	204.9, 40.6 deg
Pole RA, Dec	321.4, 66.3 deg
Length (φ1, 2–98%)	25.6 deg
Width σ(φ2), robust	1.22 deg
KINEMATICS	
pmRA* med ± σ	16.71 ± 1.03 mas/yr
pmDE med ± σ	-12.82 ± 1.40 mas/yr
μφ1* med ± σ	16.33 ± 1.09 mas/yr
μφ2 med ± σ	-13.07 ± 0.94 mas/yr
Parallax (cat.) med	0.195 ± 0.182 mas
Parallax (wtd, ZP-corr.)	0.238 ± 0.027 mas
Distance	d = 4.2 ± 0.5 kpc
Physical length	1.87 kpc
RV median ± σ(MAD)	31.6 ± 23.5 km/s
RV gradient dV/dφ1	4.15 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.1–19.9 (18.5)
(BP−RP)0 median	0.87
GSP-Phot [M/H] (n=22)	-2.17 ± 0.62
GSP-Phot Teff median	5734 K
RUWE > 1.4	0.0 %
PM clusters (largest)	2 (17 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	37 | 37
n_RV catalog | paper	11 | 11
<[Fe/H]> paper	-2.23
Stellar mass (paper)	110 Msun
Pole offset vs paper	1.3 deg
Remark	Velocities unclear

Sky map · catalog members (n = 37) · size ∝ brightness · ring = has RV

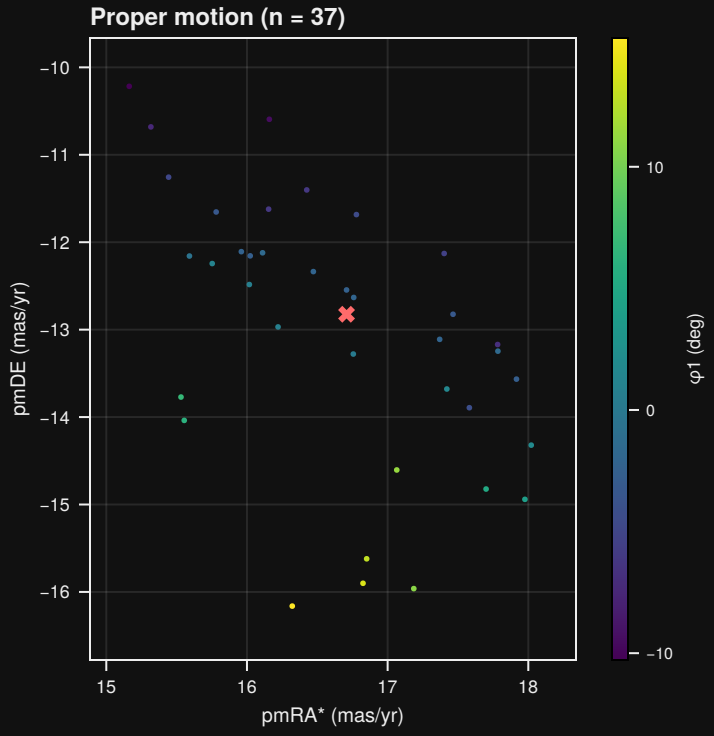


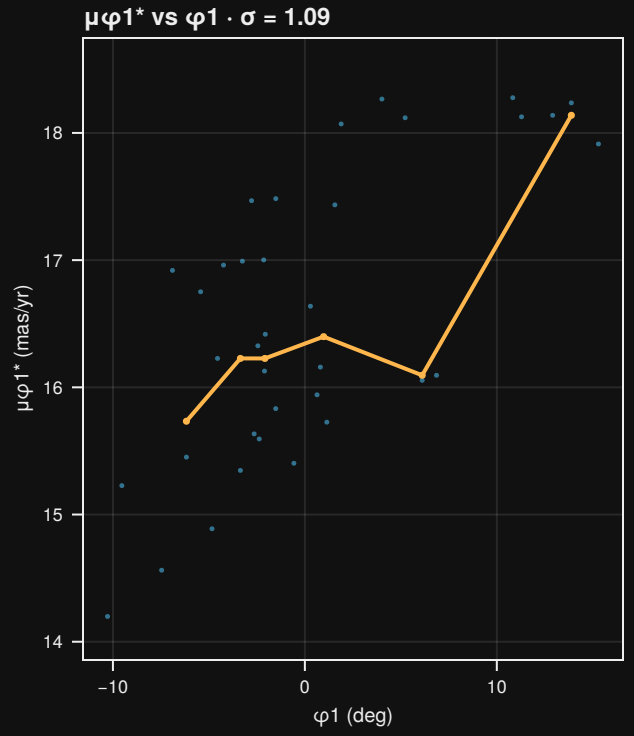
Stream frame (great-circle fit) · length 25.6° · width σ 1.22°

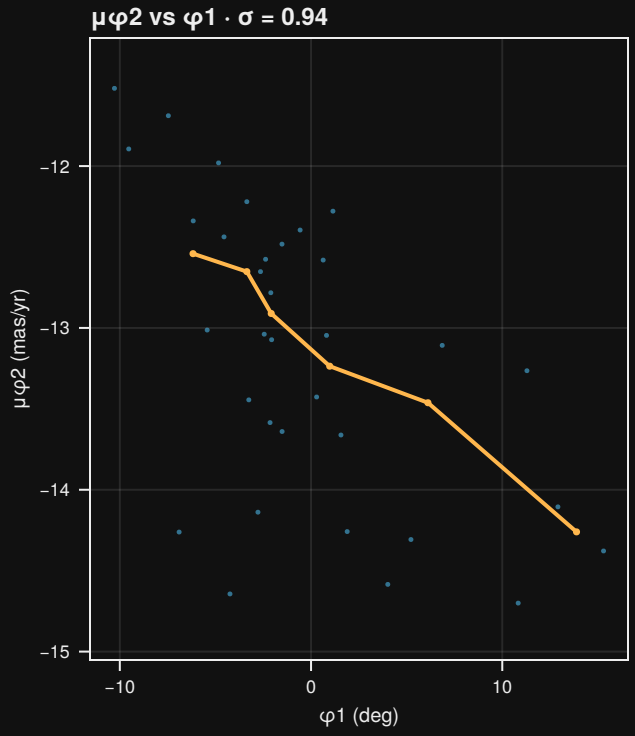


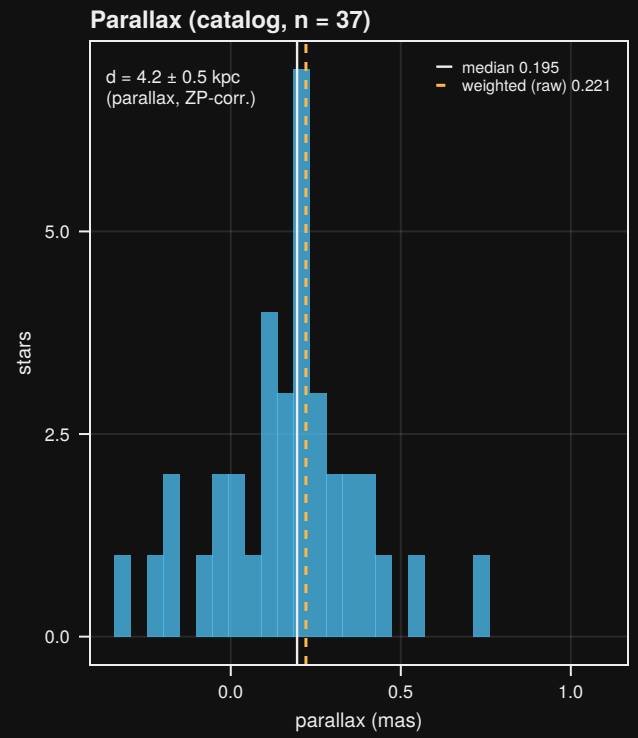
New-7

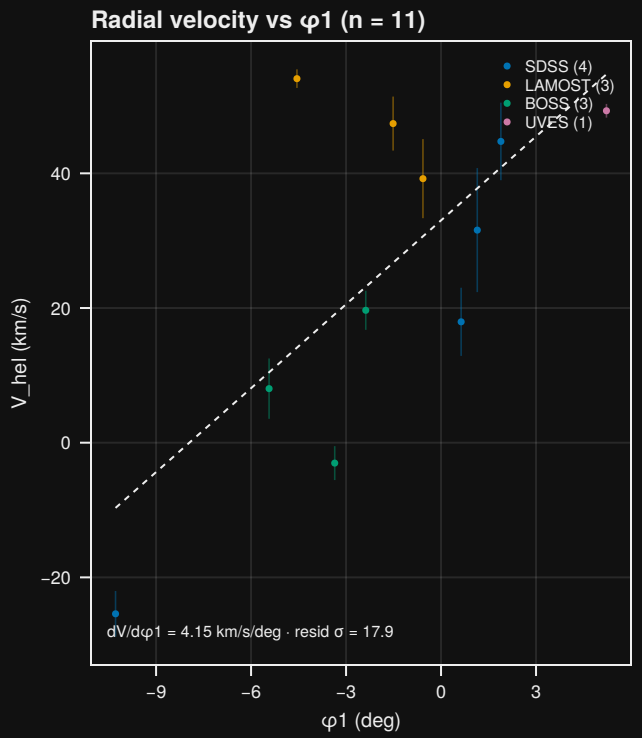
Kinematics · proper motion · parallax · radial velocity

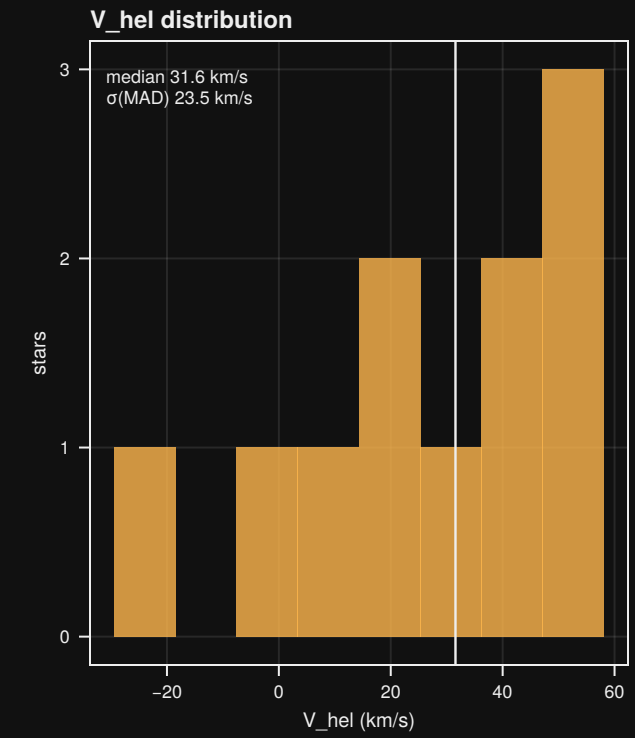








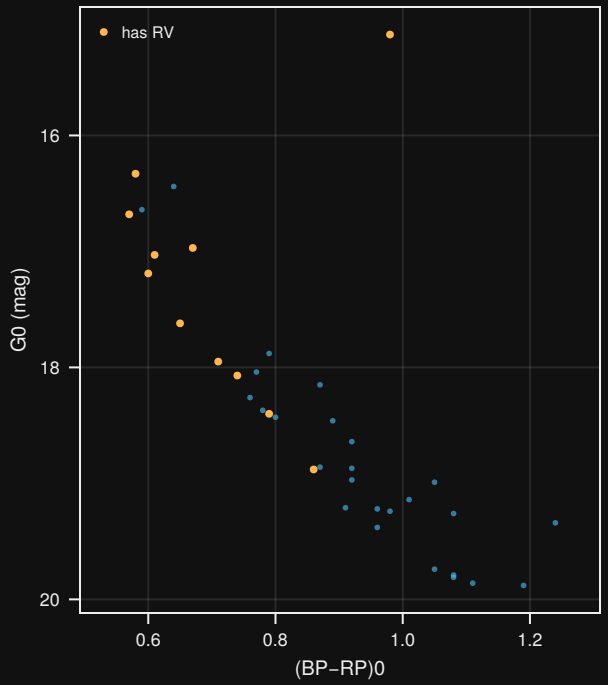




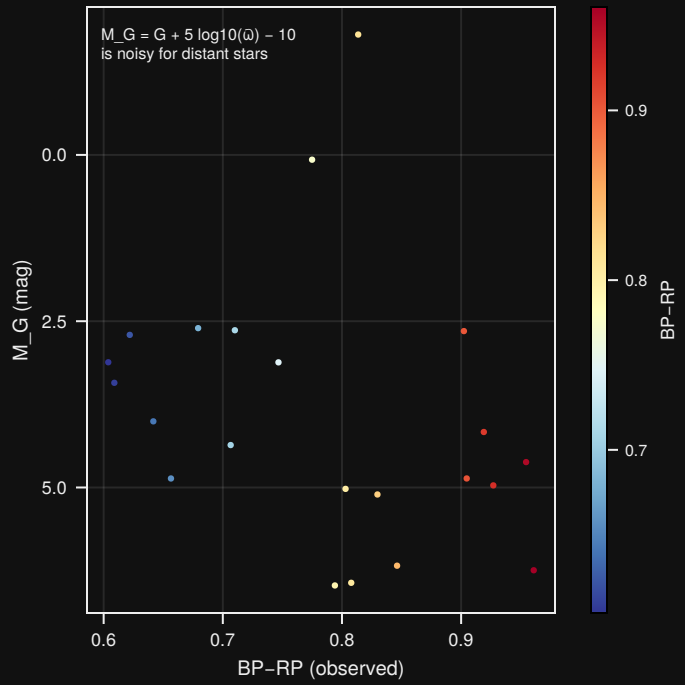
New-7

Photometry · stellar types

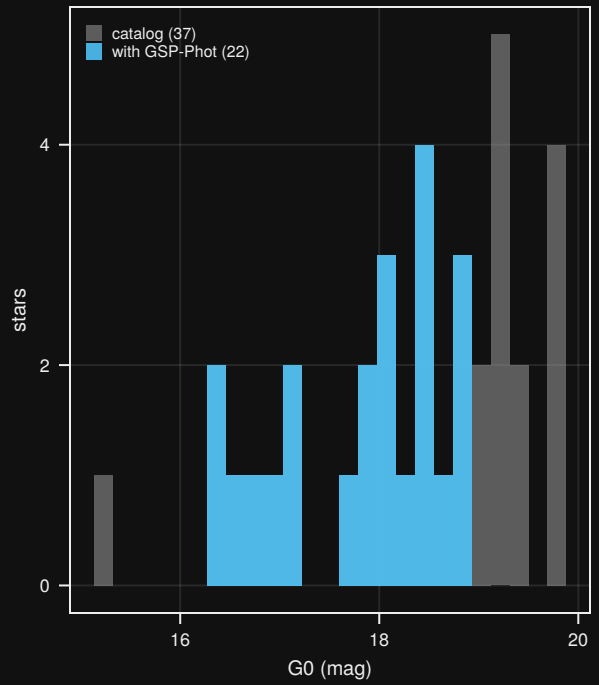
CMD · dereddened (n = 37)



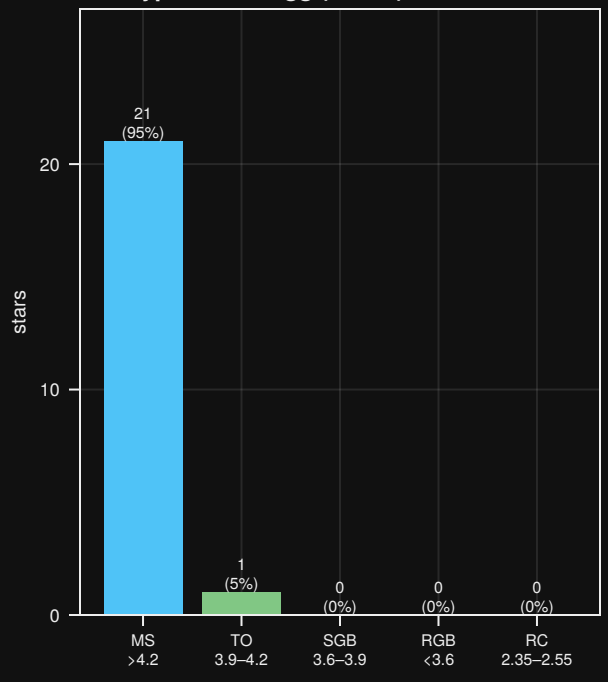
HR diagram · ϖ>0 (n = 22/22)



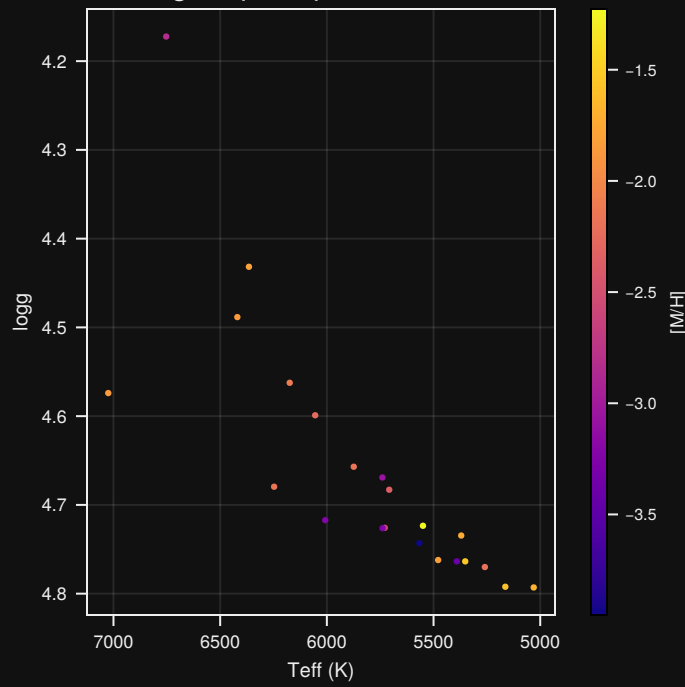
G0 & GSP-Phot coverage



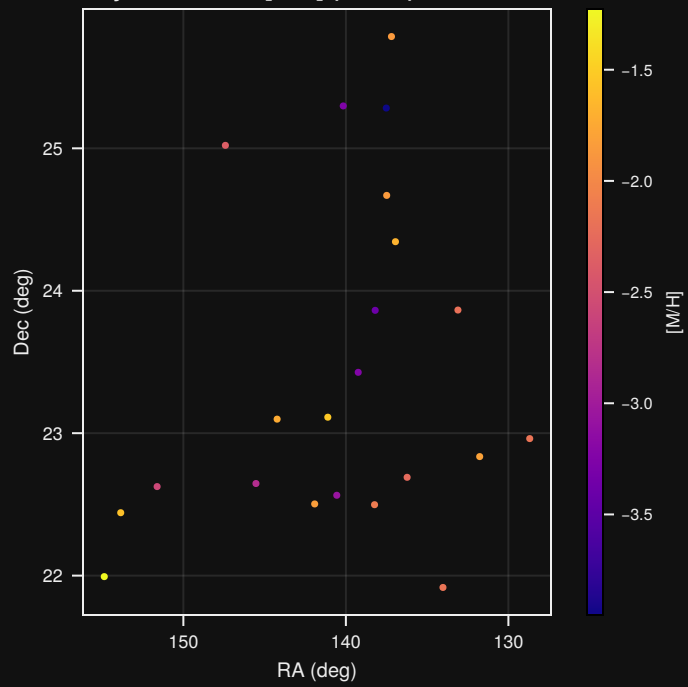
Star types from logg (n = 22)



Kiel diagram (n = 22)



Sky · GSP-Phot [M/H] (n = 22)

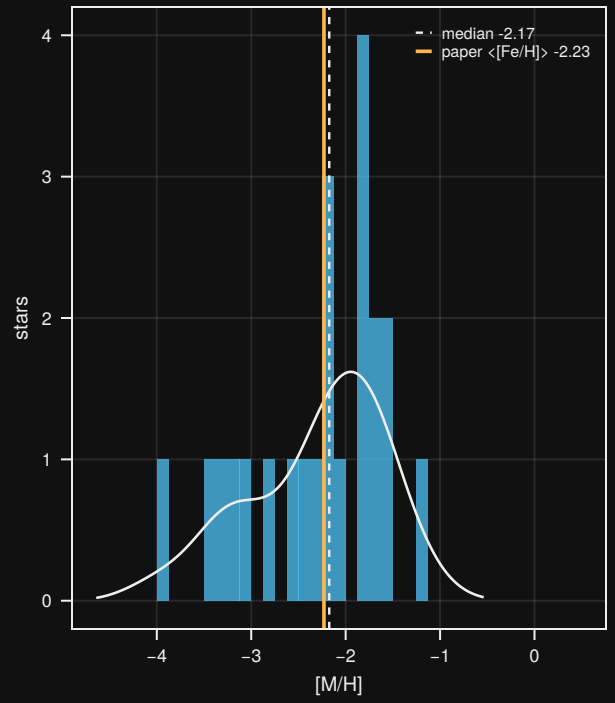


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

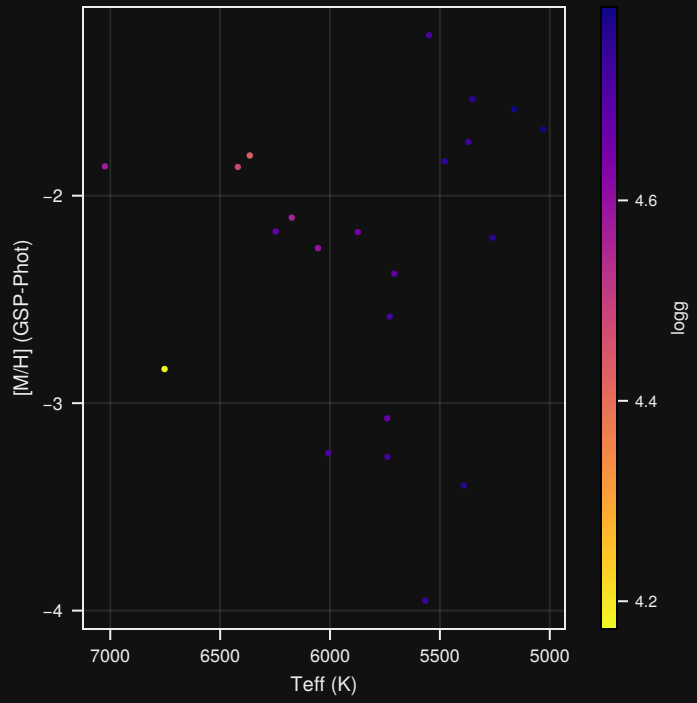
New-7

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 22)



Teff vs [M/H] (n = 22)



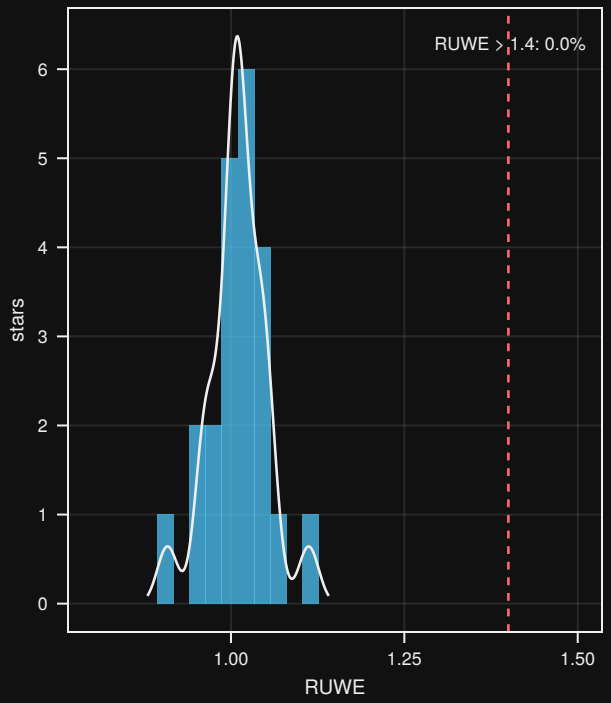
FLAME age (n = 0)

FLAME ages for only 0 star(s)
(FLAME needs good parallaxes)

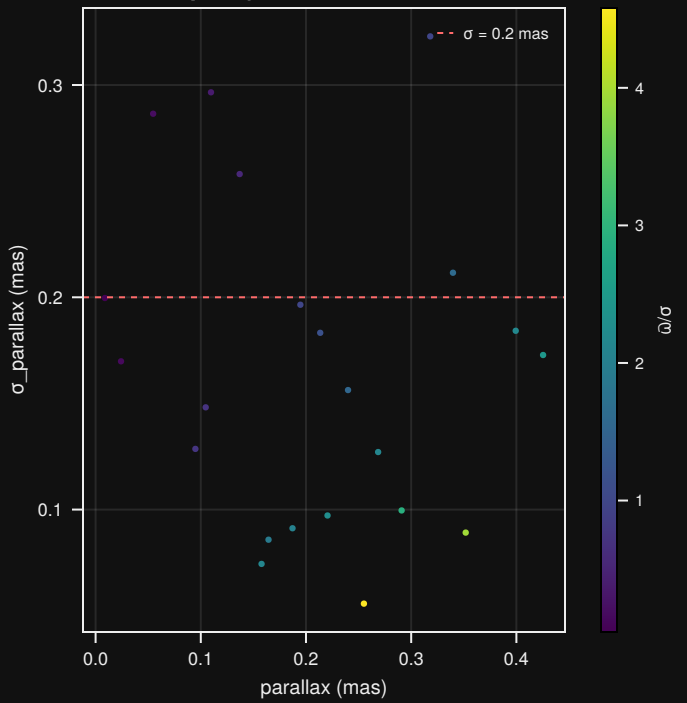
Age–metallicity (n = 0)

n < 3

RUWE (n = 22)

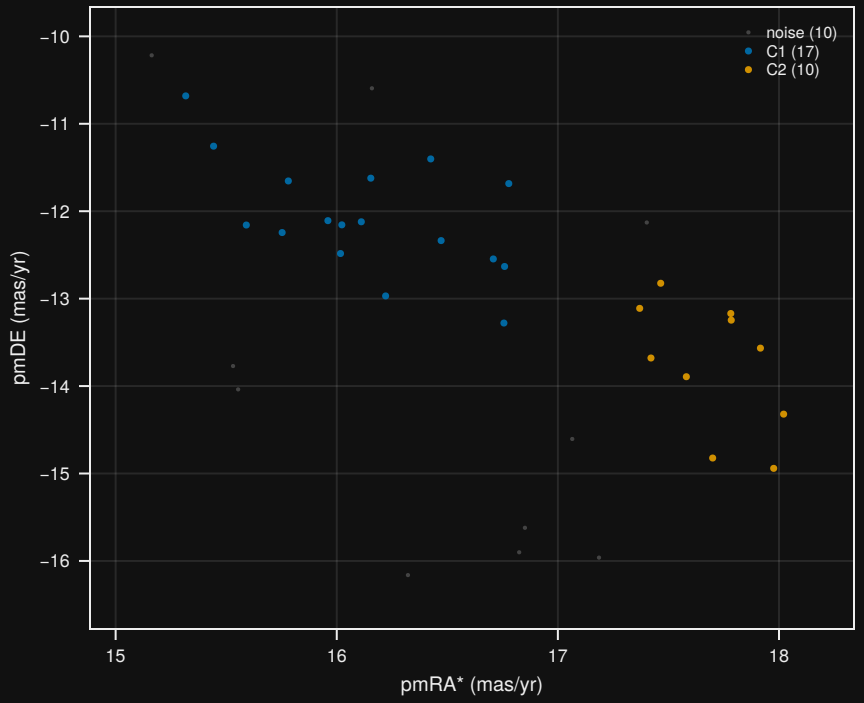


Parallax quality (n = 22)

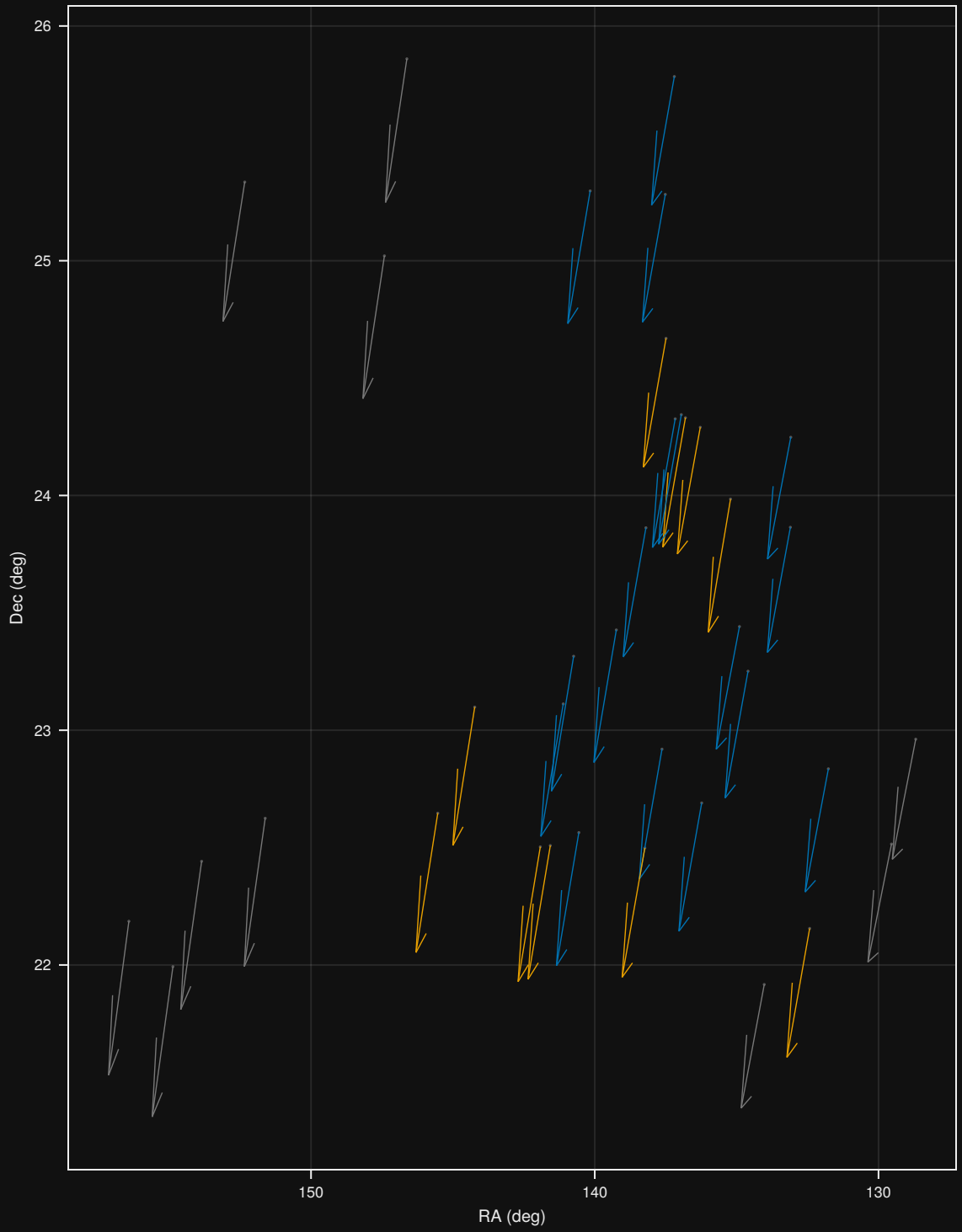


New-7

PM clustering · HDBSCAN · 2 cluster(s) · noise 27%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	10	16.57	-14.32	0.82	-2.17	2
C1	17	16.11	-12.16	0.53	-2.25	6
C2	10	17.74	-13.62	0.30	-1.86	3

New-7

Per-cluster proper-motion vector maps

